Lecture 4: AI in Medical Fields

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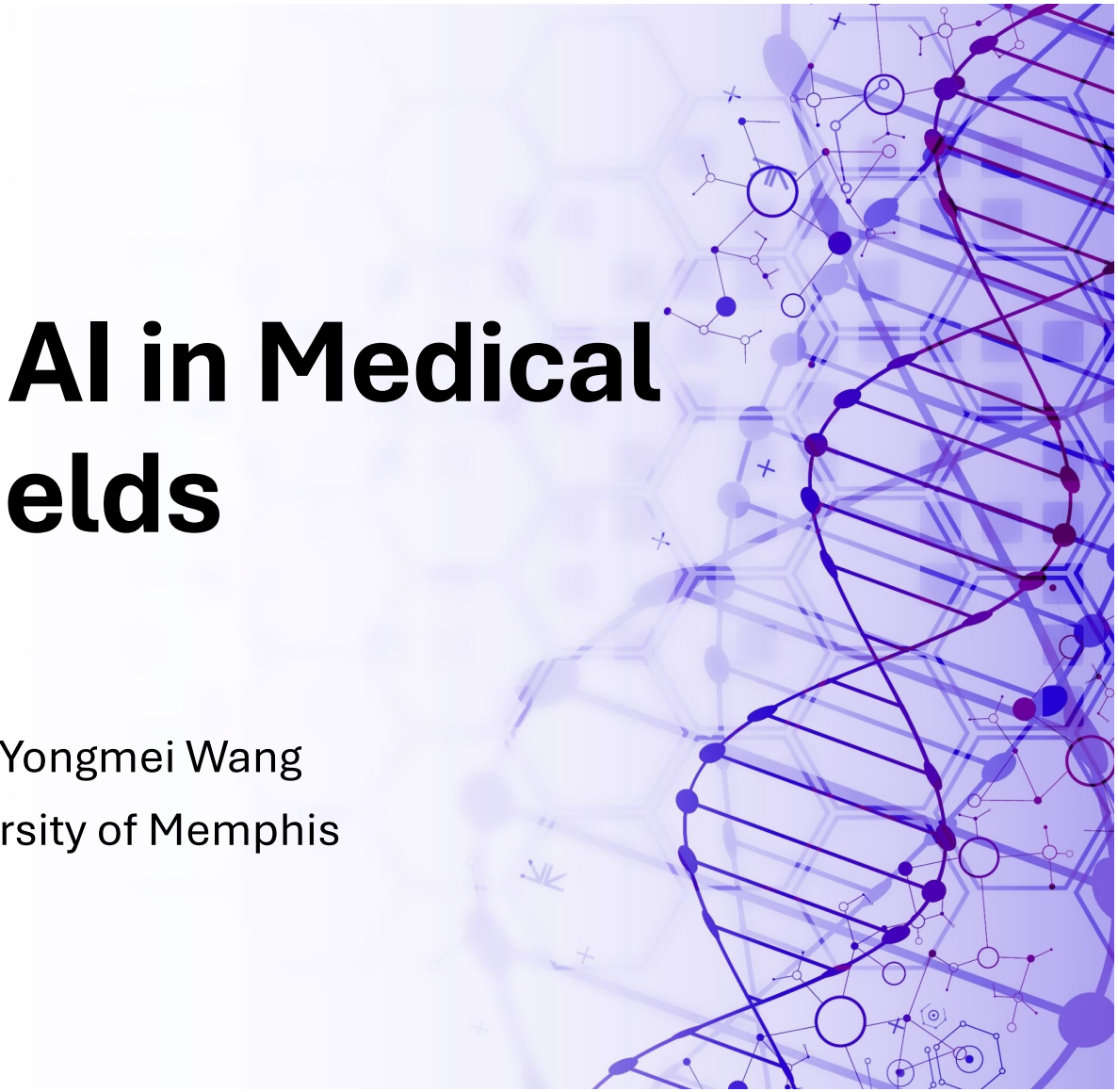


What does AI do?

Classification

Apples

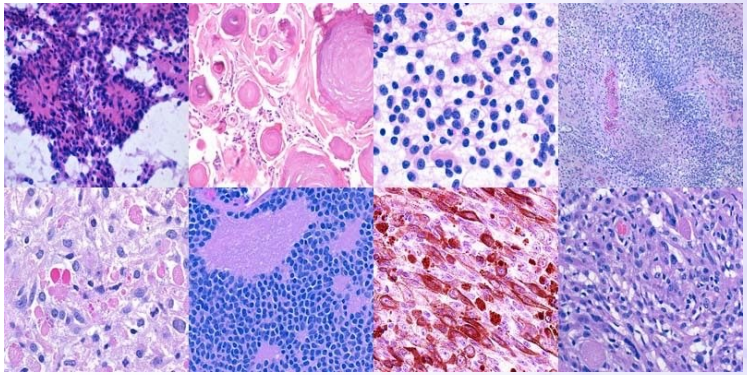
Pears

Orange

Peach

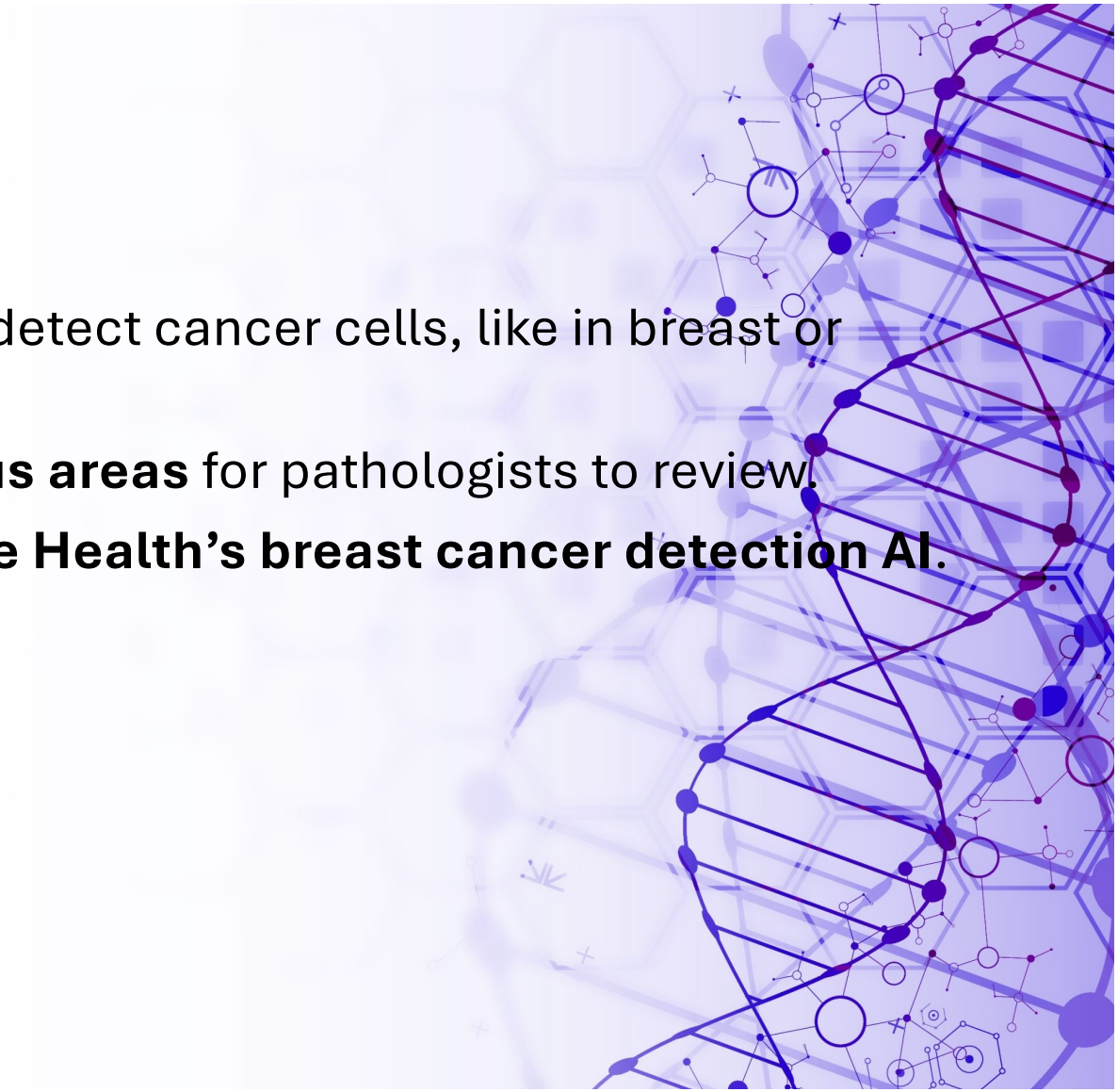
Example of use of AI : Medical Pathology

- Pathologists look at **biopsies** (tiny pieces of tissue) to check for diseases like **cancer**.
- AI, once trained, can do a better job than an expert pathologist!
- AI's job is to tell if a sample is "Benign" or "Cancerous"



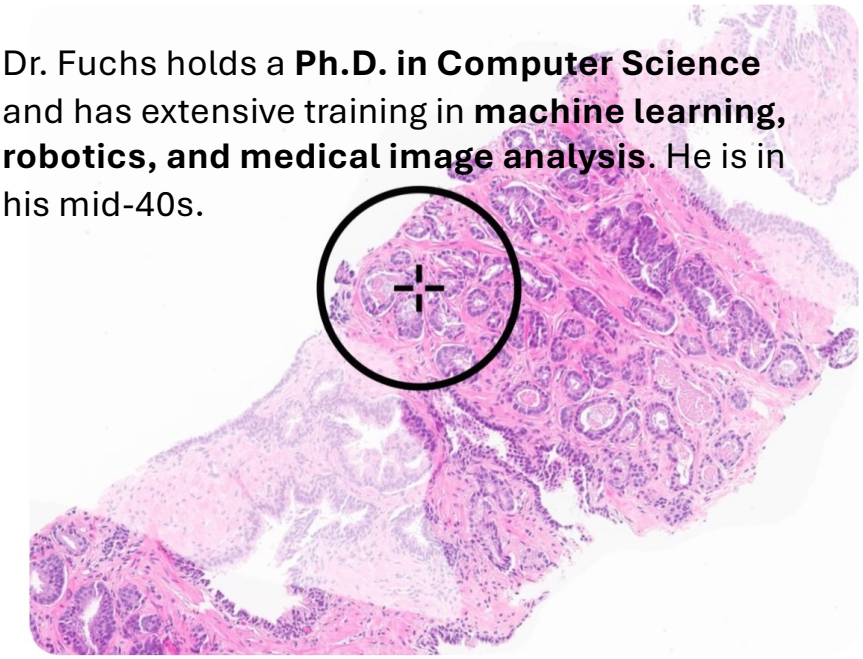
AI in pathology

- **AI reads digital slides** to detect cancer cells, like in breast or prostate cancer.
- Helps **highlight suspicious areas** for pathologists to review.
- Example: **Paige.AI**, **Google Health's breast cancer detection AI**.



PAIGE.AI company:
2017: Founded by Dr. Thomas Fuchs and colleagues.
2019: Received **FDA Breakthrough Device designation** for its prostate cancer detection system.
2021: Became the **first AI pathology product to receive FDA approval** for use in clinical diagnosis (Paige Prostate).

Dr. Fuchs holds a **Ph.D. in Computer Science** and has extensive training in **machine learning, robotics, and medical image analysis**. He is in his mid-40s.



https://www.paige.ai/diagnostic-ai

Paige Prostate

The Paige Prostate Suite is a group of comprehensive AI-powered applications that aid in the detection and grading of prostate cancer on H&E-stained whole-slide images of prostate needle biopsies.

Paige Prostate Detect was the first AI-based software application in pathology to receive **FDA approval** to aid in the primary diagnosis of prostate cancer.[ii]

Paige Prostate Detect[ii]

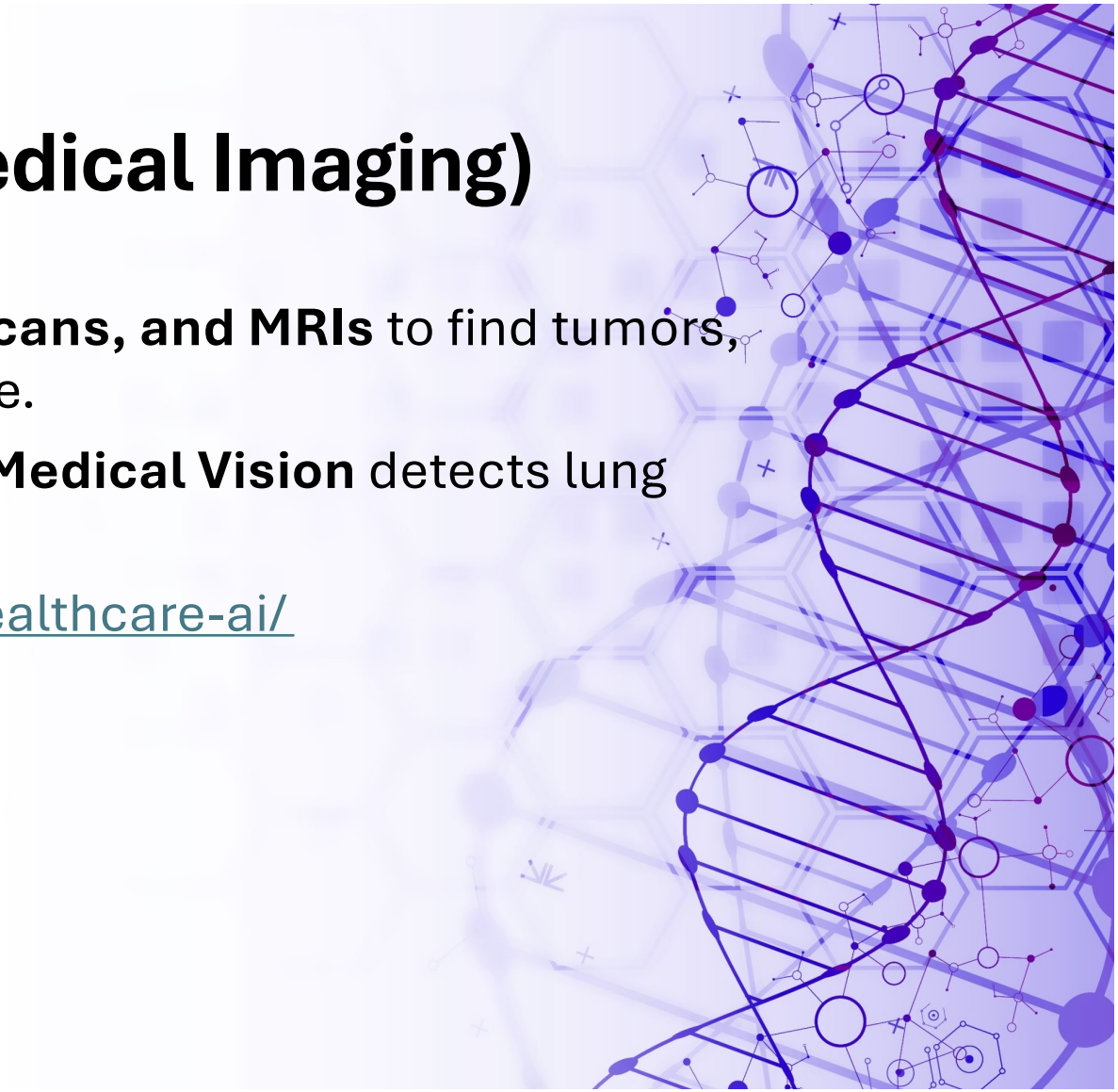
Paige Prostate Grade & Quantify[iii]

Paige Prostate Perineural Invasion (PNI)[iii]

AI in Radiology (**Medical Imaging)**

- AI helps read **X-rays, CT scans, and MRIs** to find tumors, fractures, or signs of stroke.
- Example: **Aidoc** or **Zebra Medical Vision** detects lung nodules or brain bleeds.

https://www.aidoc.com/healthcare-ai/



Other areas of AI

- **AI in Genomics & Personalized Medicine**
  - **Example: Helix, TEMPUS etc.**
- **AI as Surgical Assistance**
  - AI-powered robots (like **Da Vinci Surgical System**) help surgeons do precise operations.
  - Some AI tools analyze surgical videos to guide doctors in real-time.

https://www.intuitive.com/en-us/products-and-services/da-vinci

Major areas of AI (Fields/practices)

Area	Description
🤖 **Machine Learning**	Algorithms that learn from data
🧠 **Deep Learning**	Neural networks that handle images, speech, etc.
🔍 **Computer Vision**	Understanding images and videos
🗣️ **Natural Language Processing (NLP)**	Understanding and generating human language
📊 **Reinforcement Learning**	Training agents through trial and error
📡 **Multimodal AI**	Combining text + images + audio + video
🛡️ **AI Ethics & Safety**	Making AI fair, responsible, and secure
🔬 **Explainable AI (XAI)**	Making AI decisions understandable

AI model needs to be trained first (machine learning)

- Training an AI model means **teaching it how to understand patterns** by giving it **lots of examples** (data).
- It's kind of like how we learn: the more examples we see, the better we get.

- To train an AI to spot **cancer in pathology slides**, you feed it **thousands of labeled images**:
- "Here's what prostate cancer looks like" "Here's normal tissue"
- It learns to **recognize patterns** linked to disease.

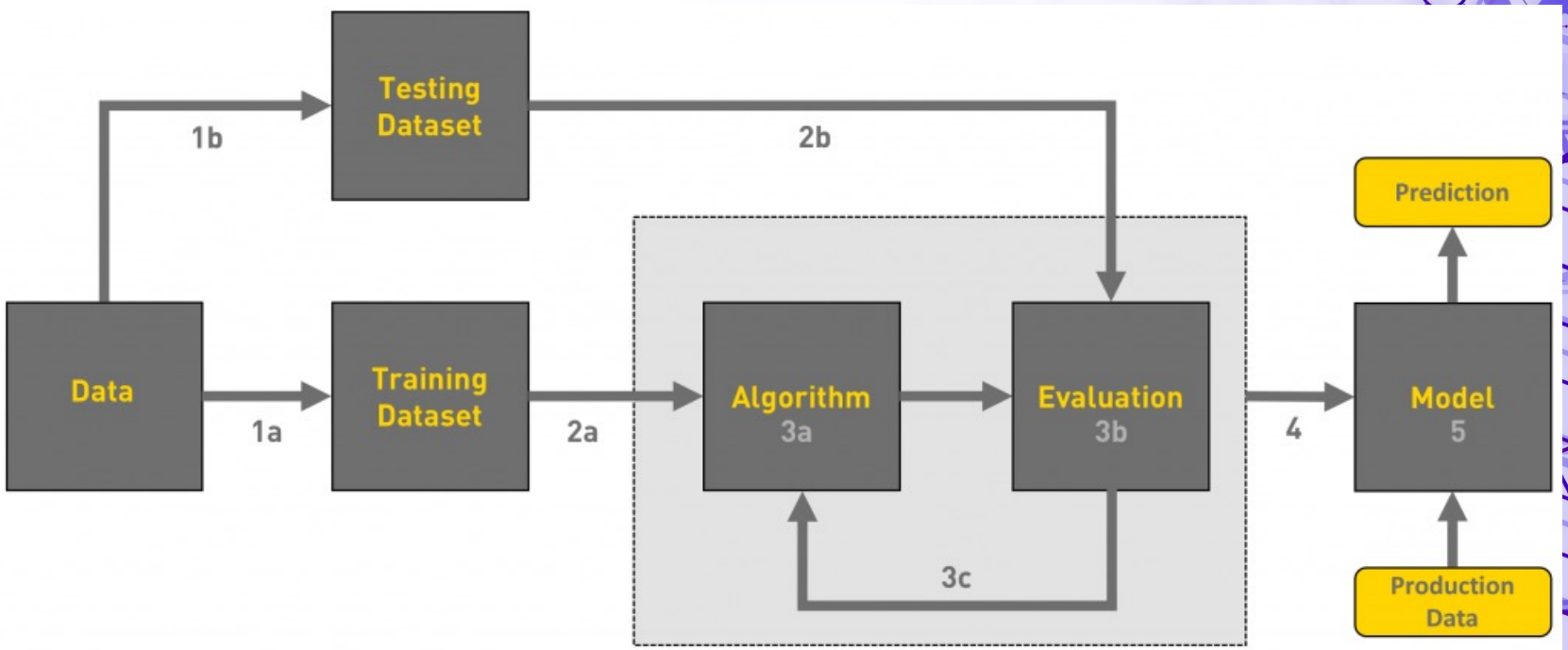
Types of Training (machine learning)

Type of AI Training	What It Needs
Supervised Learning	Labeled data (e.g. “This is cancer”)
Unsupervised Learning	Just data, no labels (finds patterns itself)
Reinforcement Learning	Learns by trial and error (like a game)

Scikit-Learn (sklearn)

- Visit Scikit-learn (also called sklearn) website first, and check to see all the modules
- Three major classes of machine learning: **Classification, Regression, and Clustering**
  - If you're trying to create a machine learning model to predict whether an email is spam or not spam, you're working on a classification problem (whether something is one thing or another)
  - If you're trying to create a machine learning model to predict the price of houses given their characteristics, you're working on a regression problem (predicting a number)
  - Or, if you're trying to get a machine learning algorithm to group together similar samples (that you don't necessarily know which should go together), you're working on a clustering problem

General workflow of Machine Learning



https://medium.com/data-science/workflow-of-a-machine-learning-project-ec1dba419b94

This website is a good introduction to some concepts in machine learning . Please visit this site to learn more.

In-class activity

- Take our data downloaded last time, GTEX_Integrin_7_organs.xlxs
- Each sample has 27 data points (27 integrin expressions)
- The label “organs” tell us that row of data is from a specific organ type

- We will use this dataset to create a ML model using one Integrin expression only, then two integrin expressions for binary classification (brain versus lung)
- We will use this dataset to create multiclass classification with selected integrin genes

How to evaluate the performance of a trained ML model?

- There are several metrics we use to evaluate the performance of a trained ML model: Accuracy, Precision, Recall (aka Sensitivity), F1 scores
- **Accuracy** is the first metric. Accuracy is defined as the Proportion of correctly predicted samples. Accuracy is a **good indicator when data is balanced, but it is bad when one set is much greater than the others**
- **Precision** is defined as the number of predicted positive labels that were actually correct. It is a **good metric** when **False positives** are costly. For example: False positives in medical tests can lead to unnecessary further testing, anxiety for patients, and increased healthcare costs.

Recall (aka Sensitivity or True Positive Rate)

- This measures how many actual positives the model caught.
- **Good when**: **False negatives** are costly (e.g., cancer detection).

F1 Score

- **Definition**: Harmonic mean of precision and recall.
- **Good when**: You want a balance between precision and recall.

Confusion Matrix: Shows actual vs predicted labels as a 2×2 table (in case of multiclass classification, it can be n× n table)

Confusion Matrix

A perfect model would have all off-diagonal elements zero.

Precision=1, Sensitivity=1, Specificity=1

Total tested samples
= TP+FN+FP+TN
Sum of all four numbers

Total Positive samples = TP+FN
Total Negative samples = FP+ TN

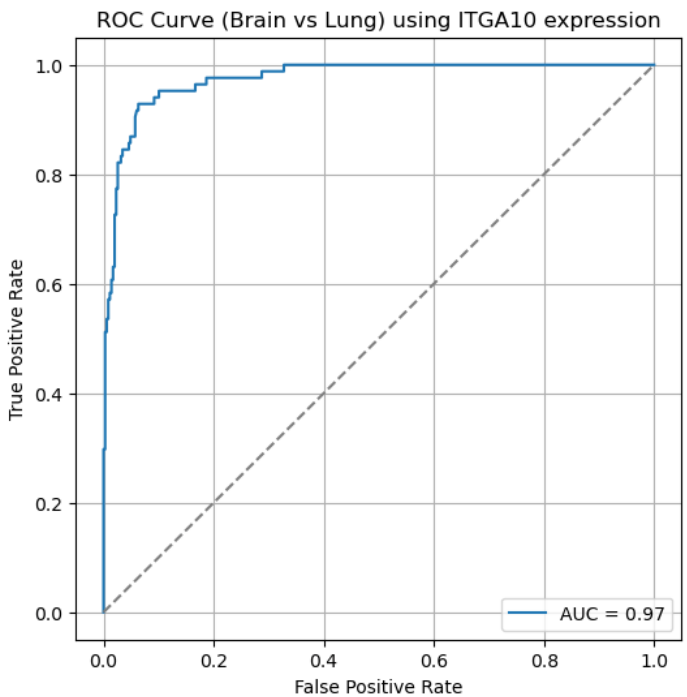
		Predicted Class: Positive	Predicted Class: Negative	
Actual Class	**Positive**	True Positive (TP)	False Negative (FN) **Type II Error**	**Sensitivity** $\frac{TP}{(TP+FN)}$
	Negative	False Positive (FP) **Type I Error**	True Negative (TN)	**Specificity** $\frac{TN}{(TN+FP)}$
		Precision $\frac{TP}{(TP+FP)}$	**Negative Predictive Value** $\frac{TN}{(TN+FN)}$	**Accuracy** $\frac{TP+TN}{(TP+TN+FP+FN)}$

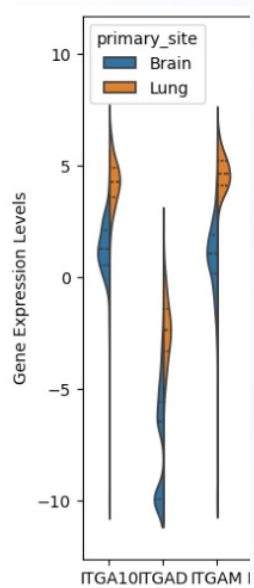
$$F1 = 2 \times \frac{Precision \times Sensitivity}{Precision + Sensitivity}$$

AUC ROC curve

- Precision, Accuracy, and F1 score report the model prediction results at a determined threshold. Their values are “threshold-dependent”.
- A ROC (Receiver Operating Curve) is **continuous prediction probabilities**, not hard labels. It evaluates the model’s **ranking ability** — how well it assigns **higher probabilities** to the correct class
- A ROC curve is obtained when one plots true positive rate (TP/(TP+FN)) against False Positive Rate (FP/(TN+FP) (which is also (1- specificity)) at various threshold settings
- AUC (Area under the curve) =1.0 means the model has excellent performance

Examples of AUROC CURVE





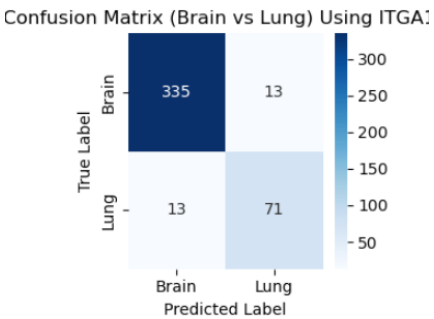
We used logistic regression. Logistic regression use the following equation

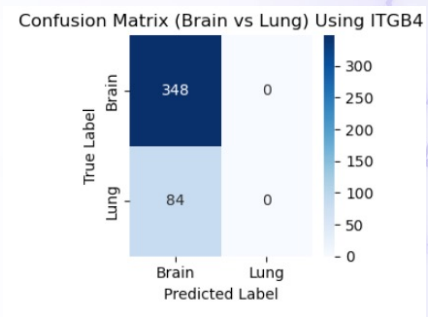
P = 1 / (1 + exp(-(a * ITGA10 + b)))
The threshold for ITGA10 to give the best prediction: 3.53

This ROC curve is generated using Gtex_expression of integrins for Brain and Lung
Brain sample: 1152, Lung sample is 288. ITGA10 expression is lower in brain than in lung.

Homework Assignment

- When we did binary classification between brain versus lung using ITGB4 expression, we got an AUROC=0.47, but our accuracy =0.82, which seems OK. Why is that so? Is accuracy a good measure in this case? If not, explain why it is not
- Here are the two confusion matrices printed out for using ITGA10 and ITGB4. Use the number shown, and calculate the following quantities: Accuracy, Sensitivity, Specificity, and F1 score manually!





Continued

- Continue from today's Jupyter notebook, but choose the other two tissues, for example lung and liver
- Then do a split violin plots of the all the integrin expression that compares lung and liver
- Then do a binary classification between lung and liver by choosing one integrin that can give you the best performance and one integrin that will give you the worst performance. Generate two ROC curves in both cases. Also get the confusion matrix and model accuracy.